SURVEYING & LEVELLING · NOTE SL02

Linear Measurement — Pacing, Chains, Tapes, Ranging and Chaining

Course topics: SWCE1.4 L3

Horizontal distance

Surveyors always record the distance on a horizontal plane. Reduce any sloping distance to its horizontal equivalent.

Direct methods	Indirect methods
Pacing, chains, tapes	Stadia method, aerial photographs

Pacing

1. Walk a known distance — usually 100 m — and count the steps.
2. Divide the distance by the steps to get your **pace factor** (average length of one step, in cm). A normal walking pace is about 80 cm (ICAR eCourse).
3. Multiply the steps counted on a new line by the pace factor.

- Accuracy is approximate; an error up to ±2% is acceptable.
- The pace changes with the person, the roughness of the ground and the slope.

Chains

Chain	Length and links	Markings
Surveyor's chain	Galvanised iron wire; 100 or 150 links; one link = distance between centres of two consecutive middle rings	Brass handles at both ends; brass tags at distinctive points
Metric chain	20 m and 30 m; each link = 20 cm	Brass tags every 5 m; rings every metre
Engineer's chain	100 ft; 100 links of 1 ft each; distances booked in chains and feet	Tags every 10 links; the centre tag is circular
Gunter's (surveyor's) chain	66 ft; 100 links of 0.66 ft (7.92 in) each	—

- 10 Gunter's chains = 1 furlong; 80 Gunter's chains = 1 mile.
- 10 square Gunter's chains = 1 acre (43,560 sq ft).

Tapes

- Tapes are more accurate than chains, graduated in cm and m, and come in lengths of 5–100 m.
- **Metallic cloth tape** — woven linen with brass or copper wires. Moderate accuracy; moisture affects it; use it where high accuracy is not needed.
- **Steel tape** — for accurate work; commonly 25 m and 50 m long. Markings are etched, riveted, or Babbitt-metal bosses. It is delicate.

- **Care of a steel tape:** avoid kinks, jerks and sharp bends; clean and oil it after use; keep it from rusting.

Auxiliary instruments

Item	Use and size
Arrows (pins)	Mark the end of each chain or tape length; stout wire, 30–40 cm long
Ranging rods	Mark points and range lines; 2–3 m long, painted alternately black and white or red and white
Plumb bob	A pointed metal weight on a string; transfers a point vertically above or below

Indirect distance measurement

- **Stadia method:** read the stadia hairs of a level or theodolite on a staff. Distance = stadia interval × 100; accuracy is about ±1%.
- **Aerial photographs:** pick two well-defined ground points, measure their distance on the photo (A) and on the ground (B). Scale of the photograph = B/A.

Ranging a survey line

Ranging fixes intermediate points on the straight line between two end stations. Do it before chaining whenever the line is long, the end stations are not clearly visible or the ground is undulating.

Direct ranging (end stations intervisible)

1. Stand behind one end station and sight the other.
2. Signal the helper holding a ranging rod to move right or left until the rod falls on your line of sight.
3. When the rod disappears behind the forward rod, fix the point.

Indirect ranging (end stations not intervisible)

- **Across a valley:** fix intermediate points alternately from opposite sides — C, then D, then E and so on — keeping the pole tops slightly above the line of sight.
- **Over a hill:** place a pole at C roughly on line AB, then D on line CA; shift C onto line DB, then D onto line CA again; repeat until A, D, C and B lie on one straight line.

Chaining

Person	Duties
Leader (head chainman)	Moves forward, drags the chain, inserts an arrow at each chain length
Follower (rear chainman)	Holds the rear end, aligns the chain, collects the arrows, keeps the record

1. The follower holds the rear end on the starting point.
2. The leader drags the chain forward; align it with the ranging rod.
3. Insert an arrow at the end of each chain length; repeat to the end station.
4. Count the arrows with the follower — that is the number of full chain lengths. Measure the remaining fraction by counting links.
5. Record each transfer of arrows in the field book; if no pegs are used, the leader keeps count of stations.

Chaining on sloping ground

- Slope below 2%: the sloping distance is acceptable.
- Slope above 2%: measure the horizontal distance.
- **Plumb bob:** hold the chain or tape horizontal and drop the end point to the ground with a plumb bob — uphill or downhill.
- **Breaking the chain (stepping):** on steep or undulating ground, measure in short horizontal steps — one end on the ground, the other held level and plumbed down — until you reach the end point.

On a long regular slope, measure the slope distance s and the difference in elevation h, and compute the horizontal distance d:

$$d = \sqrt{(s^2 - h^2)}$$

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Each link of a 20 m metric chain is

(a) 10 cm (b) 20 cm
(c) 25 cm (d) 30 cm

Q2. Length of Gunter's chain is

(a) 33 ft (b) 66 ft
(c) 100 ft (d) 20 m

Q3. 10 square Gunter's chains equal

(a) 1 hectare (b) 1 acre
(c) 1 furlong (d) 1 mile

Q4. In the stadia method, distance equals the stadia interval multiplied by

(a) 10 (b) 50
(c) 100 (d) 1000

Q5. Above what slope must the horizontal distance be measured instead of the sloping distance?

(a) 0.5% (b) 1%
(c) 2% (d) 10%

Q6. Ranging over a hill, where the end stations are not intervisible, is

(a) direct ranging (b) indirect ranging
(c) reciprocal ranging (d) stadia ranging

Answers

Q1 — (b) 20 cm
Q2 — (b) 66 ft — *100 links of 0.66 ft (7.92 in).*
Q3 — (b) 1 acre — *1 acre = 43,560 sq ft.*
Q4 — (c) 100
Q5 — (c) 2%
Q6 — (b) indirect ranging